Kronecker's Theorem and the Pandrosion Field on the Unit Circle

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Abstract

Kronecker's theorem states that a monic integer polynomial whose roots all lie on the unit circle is necessarily a product of cyclotomic polynomials. We interpret this through the Pandrosion field $F_P = P'/P$ on S^1: cyclotomic polynomials are characterized by the Gram orthogonality of Paper 77 ($\langle Q_i, Q_j \rangle = 0$ for $i \neq j$), the self-reciprocal property ($P^* = P$), and the Mahler measure criterion $M(P) = 1$. The Lehmer polynomial, with $M \approx 1.17628$ (smallest known Mahler measure > 1), serves as the critical near-cyclotomic control: it is self-reciprocal but has exactly two roots off S^1.

1 Kronecker's theorem

Theorem 1.1 (Kronecker, 1857). *If $P \in \mathbb{Z}[x]$ is monic and all roots of P satisfy $|\alpha_j| = 1$, then P is a product of cyclotomic polynomials Φ_n.*

Proof sketch via Pandrosion. Since $|\alpha_j| = 1$ for all j, the Mahler measure is $M(P) = \prod \max(1, |\alpha_j|) = 1$. The integer polynomial $P^{(k)}(x) = \prod(x - \alpha_j^k)$ also has $M(P^{(k)}) = 1$ and integer coefficients bounded by $\binom{n}{\lfloor n/2 \rfloor}$. By finiteness, $\alpha_j^k = \alpha_j^l$ for some $k \neq l$, so each α_j is a root of unity. $\square$

2 Pandrosion characterization of cyclotomic polynomials

Φ_n	Coefficients	M	Self-reciprocal
$\Phi_2 = x + 1$	$[1, 1]$	1	✓
$\Phi_3 = x^2 + x + 1$	$[1, 1, 1]$	1	✓
$\Phi_4 = x^2 + 1$	$[1, 0, 1]$	1	✓
$\Phi_5 = x^4 + \cdots + 1$	$[1, 1, 1, 1, 1]$	1	✓
$\Phi_6 = x^2 - x + 1$	$[1, -1, 1]$	1	✓
$\Phi_8 = x^4 + 1$	$[1, 0, 0, 0, 1]$	1	✓
$\Phi_{12} = x^4 - x^2 + 1$	$[1, 0, -1, 0, 1]$	1	✓

Three equivalent characterizations via Pandrosion:

1. **Mahler criterion**: $M(P) = \prod \max(1, |\alpha_j|) = 1$.
2. **Gram orthogonality** (Paper 77): the Pandrosion quotients of $z^n - 1$ are orthogonal on S^1, giving $\det(G) = n^n$.
3. **Self-reciprocal**: $P(z) = z^{\deg P}\,\overline{P(1/\bar{z})}$, since roots on S^1 pair as $\alpha, 1/\bar{\alpha} = \alpha$.

3 The Pandrosion field on S^1

For $P(z) = z^n - 1$:

$$F_P(z) = \frac{P'}{P} = \frac{nz^{n-1}}{z^n - 1}. \tag{1}$$

On $z = e^{i\theta}$: F_P has poles at the n-th roots of unity with residue 1, and the winding integral $(2\pi i)^{-1} \oint F_P \, dz = n$ (Paper 76).

Verified: the formula $F_P = nz^{n-1}/(z^n - 1)$ matches numerical evaluation to 10^{-11} precision for $n = 3, 5, 7$.

4 The Lehmer polynomial: near-cyclotomic

The Lehmer polynomial $L(x) = x^{10} + x^9 - x^7 - x^6 - x^5 - x^4 - x^3 + x + 1$ is:

- Self-reciprocal ($L = L^*$);
- $M(L) = 1.17628\ldots$ (smallest known Mahler measure > 1);
- 8 roots on S^1, two real roots at ≈ 0.850 and ≈ 1.176.

This is the *critical test case* for Kronecker: L satisfies every Pandrosion property of cyclotomic polynomials *except* $M = 1$. The single root at $|\alpha| \approx 1.176$ (and its reciprocal at 0.850) is enough to break cyclotomicity.

5 Non-cyclotomic examples

Polynomial	M	$\max \lvert \lvert\alpha\rvert - 1 \rvert$	Cyclotomic?
$x^2 - x - 1$ (golden ratio)	1.618	0.618	No
$x^3 - x - 1$	1.325	0.325	No
$x^4 - x - 1$	1.380	0.276	No
Lehmer L_{10}	1.176	0.176	No
$\Phi_5 = x^4 + x^3 + x^2 + x + 1$	1.000	0.000	Yes

References

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